

Explanation Excel tool building energy demand – dynamic - v1.5

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The structure/guide for using the tool is given in section 1.
Theoretical background is given in section 2.

1. Structure of the tool

The file consists of four sheets:

- Input Building parameters
- Results and Graphs
- Hourly Input Data
- Calculations

Sheet Input Building Parameters

In this sheet you fill in the building and occupancy parameters that have an impact on the energy demand. Fill the data in the green cells only. (You can also change the red cells but do it only consciously). Some explanation is given in column L of the Excel.

Temperature settings

Thermostat temperature setpoints for heating and cooling. The normal temperature setpoint is used when the schedule value (hourly schedule for a week starting at cell X35) for the heating/cooling is 1.0. The setback temperature setpoint is used when the schedule for the heating/cooling is 0.5. There is no heating/cooling when the schedule is 0, else heating/cooling is possible.

Geometry

Note: a façade with orientation north is defined as follows: if you are inside the building looking through a window on this façade, you are facing the north. The solar radiation incident on this façade corresponds to orientation North.

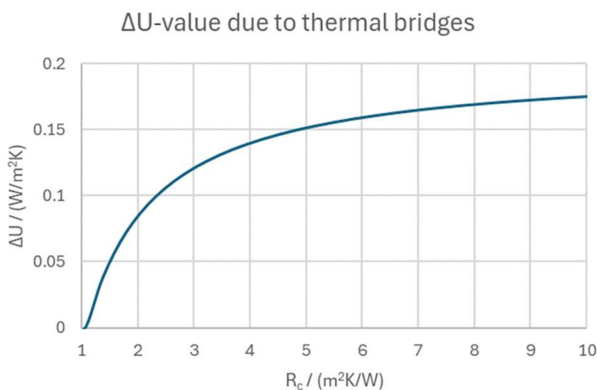
Thermal resistances and U-values

There is only one R_c -value for all walls, and all windows have the same U-value.

Thermal bridges

In addition to transmission through surface areas, there is additional transmission occurring at the junction of walls, or at the junction of wall and ceiling, wall and floor or wall and window frame, window frame and glazing. These are called thermal bridges. In guidelines a 'penalty' is sometimes given to the heat transmission coefficient (U-value) of the construction to account for these thermal bridges, especially when the exact additional heat transmission is not separately calculated or unknown. In the Dutch building energy performance standard NTA8800, the penalty ΔU is dependent on the thermal

resistance of the construction, according to the following relation:



For example, for a wall construction with a R_c -value of 4.8 m²K/W, its undisturbed U -value is $1/(4.8+0.17) = 0.20$ W/m²K. The penalty due to thermal bridges would be $\Delta U = 0.15$ W/m²K. So, the actual U -value to be used for calculating the transmission heat flow is $0.2+0.15 = 0.35$ W/m²K. The penalty thus increases the U -value by 75%.

The penalty ΔU due to thermal bridges is applied to opaque constructions, not to the windows.

Infiltration

The infiltration is given in terms of air change rate per hour (ACH). This value gives the number of times a volume of air is replaced by (in this case) outdoor air, by unintentional air leakages in the building envelope.

Ventilation

The ventilation (intentional supply of air with a quality equal to outside air) is given in air flow rate (m³/h) per person. The maximum ventilation flow rate is calculated based on maximum number of people in the building. Subsequently, this maximum ventilation flow rate is multiplied by an hourly schedule value to obtain the actual ventilation flow rate in the calculation. The schedule values are given starting at cell X7.

Remark: when the schedule for ventilation is an exact copy of the schedule for the occupancy, the ventilation is perfectly demand controlled. That means, it is exactly aligned with the occupancy. Note however that some ventilation will be desirable (a lower limit thus applies) when the occupancy is zero.

The additional ventilation rate by operable windows and night-ventilation devices can be given, and a corresponding indoor setpoint temperature. The additional ventilation with outside air is applied when:

- 1) indoor temperature is higher than this setpoint for ventilative cooling, AND
- 2) indoor temperature is higher than the setpoint temperature for heating, AND
- 3) indoor temperature is higher than outside temperature

The heat recovery (see ventilation system properties), if applicable, is bypassed when these conditions are met.

Remarks:

- the additional ventilation for cooling depends in most cases on user behavior.
- During the night, free ventilative cooling is most effective, as outside temperatures are lowest. Note that people may not fully open windows in the night due to various

reasons (safety, noise, insects). Special devices for ventilative cooling are on the market, e.g. <https://renson.net/gd-gb/products/ventilation/night-cooling>
<https://www.duco.eu/en/products/intensive-ventilation/wall-and-window-louvres/ducogrille-nightvent/ducogrille-nightvent>

Solar coefficients

Coefficients for glazing and shading/blinds can be given separately for each orientation. Note that the coefficient for shading/blinds applies to the shading system only, so it excludes the coefficient of the glazing.

Typical g_{gl} values for glazing

Single glazing	Double glazing, uncoated	Double glazing, reflective	Double glazing, low-e	Triple glazing
~ 0.8	~ 0.7	~ 0.17	~ 0.3-0.6	~ 0.4-0.6

Estimates of g_{sh} values for window shading devices

No shade	External shade	Drapery/ roller inside	Louvre half open inside	Louvre closed inside
1	~ 0.2	0.6-0.9	~ 0.7	~ 0.4

Solar absorption coefficients for outside opaque surfaces can be given as well (black is about 0.9, white is about 0.3). Note that solar heat gains by absorption and the subsequent transmission through opaque constructions is dealt with in a simplified manner where heat storage in the construction is disregarded.

Remark: if there is no glazing on a certain orientation, it does not matter which value is filled in for this orientation.

Solar blinds control

Blinds/shading are in use when

- 1) The incident solar radiation on the outside surface is higher than the threshold value
- 2) The indoor temperature is higher than the setpoint value.

Otherwise, the blinds/shading are not in use

Internal heat gains

The maximum number of persons per m² floor area is given. This maximum occupancy is multiplied by an hourly schedule value to obtain the actual occupancy in the calculation. The schedule values are given starting at cell O7.

The given maximum power density for lighting and appliances is multiplied by an hourly schedule value to obtain the actual power density in the calculation. The schedule values are given starting at cell O35.

Ventilation system properties

The system pressure drop depends on the type of ventilation system. See also comments in column L for typical values.

In case of natural ventilation, or solely mechanical supply or solely mechanical exhaust, there is no heat recovery possible and *fill in 0* for the heat recovery efficiency. If case of a mechanical supply + exhaust ventilation system with heat recovery, give the efficiency of the heat recovery.

The efficiency of a modern fan + motor system (how well electrical input is converted into air flow) is about 0.6.

Heating and cooling capacity

Give the maximum power that the heat generator and the cooling machine can produce.

Thermal capacity

The thermal capacity of the building depends on the construction characteristics. Following guidelines of the Dutch energy performance standard:

Specific thermal capacity in 1000 J/(m ² K) (m ² refers to floor area)			
		closed or lowered ceiling	no or open ceiling
Floors	Walls		
light	light	55	80
light	heavy	110	180
heavy	light	110	180
very heavy	light	110	180
heavy	heavy	180	360
light	very heavy	180	360
heavy	very heavy	250	450
very heavy	heavy	250	450
very heavy	very heavy	250	450

Floors:

very heavy: solid concrete floors
heavy: steel-concrete floors; concrete hollow-core floors; cassette floors
light: other

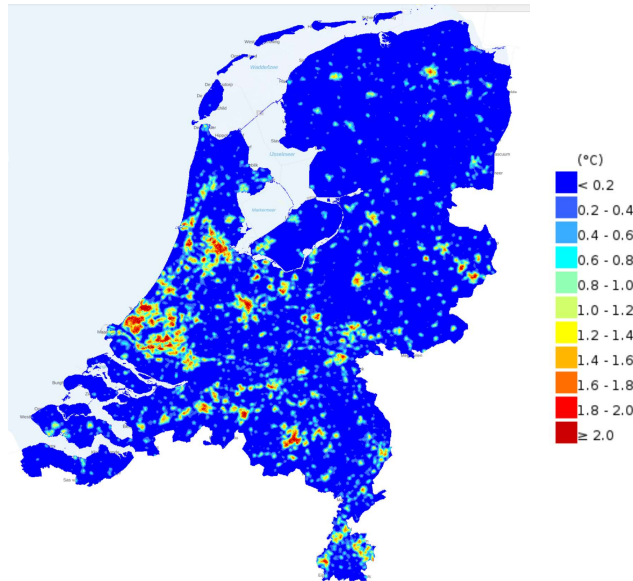
Walls:

very heavy: solid concrete walls
heavy: concrete column-beam constructions; load-bearing masonry walls
light: other

Urban Heat Island or Climate Change effect

Depending on the location of the building, the outside temperature may be somewhat elevated compared to the outside temperature in the climatic data file. Especially in cities, the temperature is higher than in the countryside. This temperature elevation in the Netherlands can be found here:

<https://tinyurl.com/ycy7v3ku>



Schedules

For every hour of a week, the schedule values used in the calculation for occupancy, lighting+appliances, heat/cooling and ventilation are given in cell range 07:AE59.

An example schedule for residential is given in cell range AK5:BA59, and for an office in cell range BE5:BU59. One can simply copy-paste these example schedules to the schedules used in the calculation.

Sheet Results

Besides numeric results for energy demand and nominal heating and cooling capacity (maximum occurring heating and cooling power in the calculation), a few standard graphs are given. Feel free to add more yourself.

Sheet Hourly Input data

This sheet contains:

- Hourly temperature data
- Hourly humidity data
- Hourly solar radiation data
- Profiles for presence, appliances and ventilation.

The sheet contains climatic data for building energy calculations from the Dutch standard NEN 5060:2018. This data is based on the meteorological station in De Bilt. It is a moderate/cold see climate.

Temperature data

- Climatic outside air temperature
- Outside air temperature corrected for heat island effect

- Running mean outside temperature, which is used for summer comfort assessment
- Ground temperature below the building (not from climate data file).

Hourly humidity data

The relative humidity data are not used in the calculation and are only presented for your information.

Hourly solar radiation data on orientations

These data are calculated based on the climatic data for direct and diffuse radiation on a horizontal surface. For the orientations, the data are calculated assuming a vertical surface.

Profiles

For every hour, the schedule values for occupancy, lighting+appliances, heat/cooling and ventilation are calculated here based on the input.

Sheet Calculations

In Columns J to AU the calculations are performed. Each row for one hour in the year. The structure is outlined below.

Thermal heat transfer coefficients (in W/K) for:

- Transmission through the construction (windows, walls and roof)
- Infiltration
- Ventilation
- Ground floor

Heat gains (Wh) due to

- Solar radiation through glazing
- Solar radiation on opaque walls and roof
(sol-air-temperature method, disregarding thermal capacity of the construction)
- Occupancy
- Lighting
- Appliances
- Fans (simplification: 50% of fan power is assumed to be given off as heat to indoors)

The inside temperature is calculated based on a few pre-calculations assuming maximum heating capacity, no heating/cooling, and maximum cooling capacity. Based on these results and input settings, the indoor temperature and actual heating demand or cooling demand are calculated. This logic is not visible.

In the rightmost columns, calculations for the summer comfort assessment are executed according to EN 16798-1 Annex B.2.2.

Sheet Results

Besides numeric results for energy demand and nominal heating and cooling capacity (maximum occurring heating and cooling power in the calculation), a few standard graphs are given. Feel free to add more!

2. Theoretical background

To determine how much heating (and cooling) a room/building needs to be maintained at a certain temperature, it is necessary to make an inventory of all energy that flows through and into the building. For the sake of simplicity, we confine ourselves to sensible heat only but note that latent heat flows can sometimes be significant. So ‘energy’ in the forthcoming refers to sensible heat.

The calculation of (non-stationary) heat transport in a room/building under realistic outdoor climate conditions is quite complicated. Computer models have been developed for this purpose. For the design of buildings and building components it is however necessary to understand the consequences of design choices on the indoor climate and energy. This insight can often be obtained by simplifying the problem, and not so much by studying very complex models. In the following, such a simplified approach is presented.

In the following, the system considered is a building or a room delimited by the outer surface of the walls, roofs and ground floor. Heat flows in the heat balance are: transmission, ventilation, infiltration, solar gains, internal heat gains and accumulation of heat in the construction.

Transmission

Because of the temperature difference between indoor air and outdoor air, heat will flow through the constructions (external walls, windows, roof and ground floor). The transmission heat flow Q_{tr} (W) through a wall, window or roof with surface area A and heat transmission coefficient U_e (W/m²K) is modelled as:

$$Q_{tr,e}(t) = U_e A (T_e(t) - T_i(t)) \quad (1)$$

With $T_e(t)$ the outside air temperature and $T_i(t)$ the inside temperature (temperature of the control volume) at time t .

For the heat transmission through a ground floor, the temperature at the external side of the floor should be considered. That temperature is much dependent on the geometry of the building, presence of a (ventilated) crawl space among others, and varies more or less with the seasons. As a crude estimation, it can be assumed to be constant. The heat flow is then calculated as:

$$Q_{tr,gr}(t) = U_{gr} A (T_{gr}(t) - T_i(t)) \quad (2)$$

With T_{gr} the temperature below the ground floor. Note that thermal storage capacity of a construction is not taken into account here, the response of the transmission heat flow to a changing temperature is ‘instantaneous’. The thermal storage of the room/building itself is however taken into account, see later.

Ventilation

Air from outside (so-called fresh air) flows into the building/room through intended openings (e.g. windows or grilles) or systems (e.g. mechanical ventilation) conceived for this aim, and subsequently leaves the building/room through intended openings or air return grilles.

To ensure acceptable indoor air quality, fresh air is needed. The quantity of air needed per person depends on his metabolism and therefore on the activity level of the person. Generally, an air flow rate of 25 to 50 m³ per hour per person is needed. The heat flow Q_{vent} (W) associated with ventilation is calculated as

$$Q_{vent}(t) = (1 - \eta) \dot{m}_{vent} c_a (T_e(t) - T_i(t)) \quad (3)$$

with

$\dot{m}_{vent}$ mass flow rate of air [kg/s]
 c_a specific heat capacity of dry air [=1000 J/kg K]
 η heat recovery efficiency (-)

The heat recovery efficiency is a characteristic of the air heat exchanger unit of a mechanical ventilation system. Without heat recovery we have $\eta=0$, and with perfect heat recovery $\eta=1$. Generally, the efficiency of a system is between about 0.6 and 0.9.

Infiltration

Besides through intended openings, outside air also flows through cracks, joints and slits in the construction, e.g. at junctions between components like wall and window frame or roof and walls. This unintended air flow is called infiltration (into the building) and exfiltration (out of the building). Because it is uncontrolled and depends on the built quality of the construction and weather conditions (mainly wind), infiltration is complicated to model in detail and generally approached through practical rules of thumbs.

The heat flow Q_{inf} (W) associated with infiltration is calculated using

$$Q_{inf}(t) = \dot{m}_{inf} c_a (T_e(t) - T_i(t)) \quad (4)$$

where $\dot{m}_{inf}$ is the infiltration mass flow rate of air [kg/s].

Generally, estimates of infiltration are based on volume flows per square meter envelope surface area, or on volume flows per square meter floor area, or on air changes per hour (ACH). The relation between a mass air flow rate $\dot{m}_{inf}$ (kg/s) into a building/room and ACH is as follows:

$$ACH = \frac{\dot{m}_{inf}}{3600 \cdot \rho_a V_{vol}} \quad [h^{-1}] \quad (5)$$

where V_{vol} is the interior volume of the building/room in m³ and ρ_a the air density (≈ 1.2 kg/m³ at 20 °C). The monthly average value of the ACH can range between 0.02 h⁻¹ and 2 h⁻¹ depending on the location, size and level of airtightness of the building.

Solar radiation through windows

Solar radiation enters the building through the glazed part of windows (a window means glazing +

frame). This incoming solar radiation is not absorbed by the indoor air but mainly by the floor, furniture and surrounding walls. The temperature response of the constructions is evidently not instantaneous due to the thermal inertia of the construction. Generally, the thermal capacity of a building (see next section) affects both the heating demand and cooling demand. The higher the thermal mass of a building, the lower its heating demand because incoming solar energy can be temporarily stored in the construction. The cooling demand can also be lower, especially when free ventilative cooling is used to discharge excess (stored) heat during the night. In a steady-state model, however, inertial effects cannot be taken into consideration. The solar gain will be instantaneous.

The heat flow into the building/room due to solar radiation on a window is modelled as:

$$Q_{i,sol,gl}(t) = g_{gl} g_{sh} A_{gl} q_{sol,e}(t) \quad (6)$$

with

g_{gl}	glazing solar factor (-)
g_{sh}	solar shading coefficient (-)
A_{gl}	glazing surface area of the window (m ²)
$q_{sol,e}$	solar irradiance on the (unobstructed) outside surface (W/m ²)

The glazing solar factor g_{gl} is the fraction of the solar energy falling on the glazing at the outside that is transmitted into the building and eventually converted to heat. Its value can be accurately measured or calculated. Typical values are:

Note that g_{gl} -values differ for diffuse and direct radiation and depend on the angle of incidence of the solar beam. These dependencies are not considered here.

The solar shading coefficient should capture the effect of shading due to architectural components (frame, reveal, eave, canopy etc.; not addressed here) and window shading devices, for which estimates are:

The solar irradiance on an unobstructed outside surface, $q_{sol,e}$, depends on surface orientation and inclination, as well as on time and location. When the direct and diffuse radiation on a horizontal surface is known for a location, the solar irradiance on a surface with another orientation and inclination can be quite accurately calculated. Usually, both direct and diffusion radiation on a horizontal surface are given in weather data files on an hourly basis.

Solar radiation on opaque walls or roof

Solar radiation that is absorbed on the outer surface of an external non-transparent construction will increase its outside surface temperature. In steady state, this temperature increment equals:

$$\frac{a_{sol} q_{sol,e}(t)}{\alpha_e} \quad (7)$$

with

a_{sol}	solar absorption coefficient outside surface (-)
α_e	surface heat transfer coefficient with external environment (W/m ² K)

The absorbed heat is subsequently transmitted inwards through the opaque construction and can be

considered as a solar heat gain, calculated as:

$$Q_{i,sol,op}(t) = U_e A \frac{a_{sol} q_{sol}(t)}{\alpha_e} \quad (8)$$

Typical values for the solar absorption coefficient of materials can be found in the literature. The surface heat transfer coefficient α_e depends much on the situation and wind speed. A typical value for an average wind speed is 15 W/m²K. Note that the thermal capacity of a construction will cause a significant time delay for the 'solar induced' heat flow towards the inside surface. For well-insulated constructions (low U_e -value), the solar induced heat flow will be comparatively small and may be neglected.

Internal heat gains

Heat is produced inside buildings by people, by electrical appliances and by lighting. These are referred to as internal heat gains.

The human body produces heat through respiration, convection from skin to air and sweat. The quantity of heat released depends on the level of activity, the type of clothes and the temperature and humidity of the surrounding air. It is a combination of sensible and latent heat. The latent heat production (water vapour production) is much more dependent on activity level than the sensible component. Here we limit the attention to sensible heat only. Under normal circumstances the sensible heat production is about 65 W/pp.

For electrical appliances in buildings like computers, televisions, printers, servers, refrigerators, freezers, washing machines, coffee machines etc., as well as for lighting appliances, almost all electrical power consumed will eventually be released as heat in the building. The total electrical lighting power installed in a building depends on the type of lamps used and on the function of the building.

Typical yearly heat production (electrical consumption) of appliances and lighting together is roughly 36 kWh per m² floor area in offices and 30 kWh per m² floor area in dwellings. For offices the peak demand is during daytime, while for dwellings it is in the evening with possibly an additional peak around 18:00 for (electrical) cooking.

Note that the (yearly) electrical consumption for appliances and lighting can vary a lot between buildings. Electrical consumption by lighting can be reduced by high-efficiency LED lights, advanced control (occupancy, daylight) and task lighting.

All heat flows together

The heat flows to consider are:

$$Q_{tr,e}(t) + Q_{tr,gr}(t) + Q_{vent}(t) + Q_{inf}(t) + Q_{i,sol,gl}(t) + Q_{i,sol,op}(t) + Q_{i,int}(t)$$

to which we must add the heat flow $Q_{h/c}(t)$ due to heating or cooling. $Q_{h/c}(t)$ is >0 for heating and <0 for cooling.

For convenience, we bundle a few terms together:

$$Q_e(t) = Q_{tr,e}(t) + Q_{tr,gr}(t) + Q_{vent}(t) + Q_{inf}(t) \quad (9)$$

$$Q_i(t) = Q_{i,sol,gl}(t) + Q_{i,sol,op}(t) + Q_{i,int}(t) \quad (10)$$

$$Q_{h/c}(t) \quad (11)$$

The reason for doing so is that:

$Q_e(t)$	depends on the indoor temperature
$Q_i(t)$	does not depend on indoor temperature
$Q_{h/c}(t)$	may need to be determined based on setpoint temperature, see later

Accumulation/storage of heat; balance equation

A simplified *one-node* model for a room/building is followed. In such a model, the combined heat storage capacity of the room air, internal walls, ceilings, floors and furniture is represented by one thermal mass node. The temperature of this node, T_i , is assumed to represent the operative indoor temperature of the room/building.

The heat balance at time t is then written as:

$$Q_e(t) + Q_i(t) + Q_{h/c}(t) - M \frac{dT_i(t)}{dt} = 0 \quad (\text{J/s} = \text{W}) \quad (12)$$

with the total thermal mass M (J/K) of the room/building. It is the sum of component n with mass m_n and specific heat capacity c_n :

$$M = \sum_n (m_n c_n) \quad (13)$$

The components refer to the air volume, furniture, and several cm's of the walls, floor, and ceiling at the inside (often inner 6 cm is chosen):

For convenience, we can write for the heat flow that depends on the outside temperature:

$$Q_e(t) = H_e (T_e(t) - T_i(t)) + H_{gr} (T_{gr}(t) - T_i(t)) \quad (14)$$

with

$$H_e = \sum U_e A + (1 - \eta) \dot{m}_{vent} c_a + \dot{m}_{inf} c_a \quad (15)$$

where the summation is taken over the several sub surfaces of the external wall and roof (windows and opaque parts).

Calculation of the indoor temperature as a function of time

If we assume that $T_e(t)$ and $T_{gr}(t)$ are known at some fixed time point t , and assumed constant from t onwards (during a time periode Δt), the heat balance may be written as:

$$Q_i + Q_{h/c} + H_e (T_e(t) - T_i(t')) + H_{gr} (T_{gr}(t) - T_i(t')) - M \frac{dT_i(t')}{dt'} = 0 \quad (16)$$

with t' a continuous time variable and Q_i , $Q_{h/c}$, H_e , H_{gr} , M , $T_e(t)$ and $T_{gr}(t)$ constants. We are interested in how the indoor temperature T_i will develop as a function of time, from the time point t onwards.

The *exact* solution for the temperature T_i is described by the following equation:

$$T_i(t') = T_i(t) + [T_{i,eq}(t) - T_i(t)] \left[1 - \exp\left(-\frac{(H_e + H_{gr})}{M}(t' - t)\right) \right] \quad (17)$$

where

$$T_{i,eq}(t) = \frac{Q_i(t) + Q_{h/c}(t) + H_e T_e(t) + H_{gr} T_{gr}(t)}{H_e + H_{gr}} \quad (18)$$

$T_i(t)$ = the temperature T_i at fixed time point t

$T_{i,eq}(t)$ = the equilibrium temperature that would be reached after a long time ($t' \rightarrow \infty$):

For a time period Δt later, we have $t' = t + \Delta t$. For this point in time, we can thus write:

$$T_i(t + \Delta t) = T_i(t) + [T_{i,eq}(t) - T_i(t)] \left[1 - \exp\left(-\frac{(H_e + H_{gr})}{M} \Delta t\right) \right] \quad (19)$$

Implementation in the spreadsheet tool

The above approach is implemented in the Excel tool. Basically, from a weather data file, we have hourly values for the outside temperature and solar radiation on outside surfaces for a complete year, so $24 \cdot 365 = 8760$ lines/rows for each hour. An assumption is made for the ground temperature during the year (e.g. a constant or yearly sinusoidal). For each hour, the inside temperature at the next hour can be calculated using Eq. (19) with $\Delta t = 3600$ s. Note that parameters like ventilation rate (or others) may vary each hour.

Dedicated attention must be given to the thermostat control. When a thermostat in a room is active and regulates the amount of heating, and possibly cooling, a certain desired temperature is to be reached by the heating and cooling system.

One should also consider that:

- $Q_{h/c}$ can have an upper limit corresponding to the maximum heating capacity of the system
- $Q_{h/c}$ can have a lower limit corresponding to the maximum cooling capacity of the system.
- An inside setpoint temperature is defined below which heating is on, denoted as $T_{set,h}$.
- An inside setpoint temperature is defined above which cooling is on, denoted as $T_{set,c}$.
- An inside setpoint temperature is defined above which the windows are opened, denoted as $T_{set,v}$. Open windows give extra ventilation and thus cooling, provided outside temperature < inside temperature.
- H_e can thus depend on the state whether ventilative cooling is desired. Therefore, we have $H_{e,heat}$ and $H_{e,cool}$.

We furthermore distinguish three cases during the timestep Δt :

1. Active heating/cooling is required and within the timestep Δt the desired temperature is reached. The required heating/cooling power will be less than the maximum heating/cooling capacity of the system.
2. Active heating/cooling is required and within the timestep Δt the desired temperature will not be reached. The heating/cooling power will be at the maximum capacity.
3. Active heating/cooling is not required. This is a free-floating situation. Care must be taken here which value to take, $H_{e,heat}$ or $H_{e,cool}$ and additional checks are necessary with regard to the inside temperature and ventilation setpoint temperature.

Ad. 1. When the desired temperature is reached within Δt , there is no need to calculate the new temperature at $t+\Delta t$, because it will be the setpoint temperature of the thermostat (assuming optimal control). So:

$$T_i(t + \Delta t) = T_{set}(t) \quad (20)$$

The required heating or cooling is derived from:

$$T_{set} = T_i(t) + [T_{i,eq}(t) - T_i(t)] \left[1 - \exp\left(-\frac{(H_e + H_{gr})}{M} \Delta t\right) \right] \quad (21)$$

Or:

$$T_{set} = T_i(t) + \left[\frac{Q_i(t) + Q_{h/c}(t) + H_e T_e(t) + H_{gr} T_{gr}(t)}{H_e + H_{gr}} - T_i(t) \right] \left[1 - \exp\left(-\frac{(H_e + H_{gr})}{M} \Delta t\right) \right] \quad (22)$$

So

$$Q_{h/c}(t) = -Q_i(t) - H_e (T_e(t) - T_i(t)) - H_{gr} (T_{gr}(t) - T_i(t)) + \frac{(H_e + H_{gr})(T_{set} - T_i(t))}{\left[1 - \exp\left(-\frac{(H_e + H_{gr})}{M} \Delta t\right) \right]} \quad (23)$$

With $|Q_{h/c}| <$ the maximum heating/cooling capacity of the system. The heating setpoint temperature is denoted as $T_{set,h}$ and the cooling setpoint temperature as $T_{set,c}$. Obviously, we require that $T_{set,h} < T_{set,c}$.

Ad. 2. When the desired temperature is not reached within Δt , the temperature at $t+\Delta t$ is calculated using Eq. (19) and $Q_{h/c}$ will be at the maximum capacity.

Ad. 3. No heating and no cooling is necessary. If the inside temperature would drop below $T_{set,v}$ while using $H_{e,cool}$ in the equations, the inside temperature is set at $T_{set,v}$. For meaningful results, we require that

$$T_{set,h} < T_{set,v} < T_{set,c}$$

In the Excel tool, the following workflow is used at each timestep:

1. Pre-calculate inside temperature $T(t+\Delta t)$ for four situations:

- No heating and no cooling with $H_{e,heat}$, denoted as $T(t+\Delta t)_{free,heat}$ (free floating)
- No heating and no cooling with $H_{e,cool}$, denoted as $T(t+\Delta t)_{free,cool}$ (free floating)
- Maximum heating capacity with $H_{e,heat}$, denoted as $T(t+\Delta t)_{heat}$
- Maximum cooling capacity with $H_{e,cool}$, denoted as $T(t+\Delta t)_{cool}$

2. The actual inside temperature $T(t+\Delta t)$ is chosen according to the following flow of logic:

If $T(t+\Delta t)_{heat} < T_{set,h}$		then $T(t+\Delta t) = T(t+\Delta t)_{heat}$, $Q_h = Q_{h,max}$	else
If $T(t+\Delta t)_{heat} > T_{set,h}$	and $T(t+\Delta t)_{free,heat} < T_{set,h}$	then $T(t+\Delta t) = T_{set,h}$ (and calculate Q_h)	else
If $T(t+\Delta t)_{free,heat} > T_{set,h}$	and $T(t+\Delta t)_{free,heat} < T_{set,v}$	then $T(t+\Delta t) = T(t+\Delta t)_{free,heat}$ $Q_h = Q_c = 0$	else
If $T(t+\Delta t)_{free,heat} > T_{set,v}$	and $T(t+\Delta t)_{free,cool} < T_{set,v}$	then $T(t+\Delta t) = T_{set,v}$ $Q_h = Q_c = 0$	else
If $T(t+\Delta t)_{free,cool} > T_{set,v}$	and $T(t+\Delta t)_{free,cool} < T_{set,c}$	then $T(t+\Delta t) = T(t+\Delta t)_{free,cool}$ $Q_h = Q_c = 0$	else
If $T(t+\Delta t)_{cool} < T_{set,c}$	and $T(t+\Delta t)_{free,cool} > T_{set,c}$	then $T(t+\Delta t) = T_{set,c}$ (and calculate Q_c)	else
If $T(t+\Delta t)_{cool} > T_{set,c}$		then $T(t+\Delta t) = T(t+\Delta t)_{cool}$, $Q_c = Q_{c,max}$	